

MongoDB Sharding Lab – Complete Steps

A. Before Starting

1. Stop Normal MongoDB Service

Open CMD as Administrator:

```
net stop MongoDB
```

Verify:

```
sc query MongoDB
```

Expected:

```
STATE : 1 STOPPED
```

B. Create Fresh Directories

(Optional: delete old sharding folders first)

```
mkdir C:\sharding  
mkdir C:\sharding\config  
mkdir C:\sharding\shard1  
mkdir C:\sharding\shard2
```

C. Start Config Server

Open CMD Window #1

```
"C:\Program Files\MongoDB\Server\8.3\bin\mongod.exe" --configsvr --replSet  
configReplSet --dbpath C:\sharding\config --port 27019
```

Leave window open.

D. Initialize Config Replica Set

Connect using Compass:

```
mongodb://127.0.0.1:27019/?directConnection=true
```

Open shell and run:

```
rs.initiate({
```

```
_id: "configReplSet",
configsvr: true,
members: [
  { _id: 0, host: "localhost:27019" }
]
})
```

Verify:

```
rs.status()
```

E. Start Shard 1

Open CMD Window #2

```
"C:\Program Files\MongoDB\Server\8.3\bin\mongod.exe" --shardsvr --replSet
shard1ReplSet --dbpath C:\sharding\shard1 --port 27011
```

F. Initialize Shard 1

Connect:

```
mongodb://127.0.0.1:27011/?directConnection=true
```

Run:

```
rs.initiate({
  _id: "shard1ReplSet",
  members: [
    { _id: 0, host: "localhost:27011" }
  ]
})
```

G. Start Shard 2

Open CMD Window #3

```
"C:\Program Files\MongoDB\Server\8.3\bin\mongod.exe" --shardsvr --replSet
shard2ReplSet --dbpath C:\sharding\shard2 --port 27012
```

H. Initialize Shard 2

Connect:

```
mongodb://127.0.0.1:27012/?directConnection=true
```

Run:

```
rs.initiate({
  _id: "shard2ReplSet",
  members: [
    { _id: 0, host: "localhost:27012" }
  ]
})
```

I. Start mongos Router

Open CMD Window #4

```
"C:\Program Files\MongoDB\Server\8.3\bin\mongos.exe" --configdb
configReplSet/localhost:27019 --port 27017
```

Leave running.

J. Connect to mongos

Compass:

```
mongodb://127.0.0.1:27017
```

K. Add Shards

Run:

```
sh.addShard("shard1ReplSet/localhost:27011")
sh.addShard("shard2ReplSet/localhost:27012")
```

Verify:

```
sh.status()
```

L. Enable Sharding

```
sh.enableSharding("eCommerceDB")
```

M. Shard Collection

```
sh.shardCollection(
  "eCommerceDB.orders",
  { customer_id: "hashed" }
)
```

N. Insert Test Data

```
use eCommerceDB
for(let i=0;i<1000;i++){
  db.orders.insertOne({
    customer_id:"cust_"+i,
    amount:Math.random()*100
  })
}
```

O. Verify Distribution

Check:

```
sh.status()
```

or

```
db.orders.getShardDistribution()
```

Expected:

```
Shard1  ≈ 50%
Shard2  ≈ 50%
```

Shutdown Procedure

1. Stop mongos

In Window #4:

```
Ctrl + C
```

2. Stop Shard 1

In Window #2:

```
Ctrl + C
```

3. Stop Shard 2

In Window #3:

```
Ctrl + C
```

4. Stop Config Server

In Window #1:

```
Ctrl + C
```

5. Verify All Stopped

```
netstat -ano | findstr 27017
netstat -ano | findstr 27019
netstat -ano | findstr 27011
netstat -ano | findstr 27012
```

No LISTENING entries should remain.

Return to Normal MongoDB

Start the Windows service again:

```
net start MongoDB
```

Verify:

```
sc query MongoDB
```

Expected:

```
STATE : 4 RUNNING
```

Check:

```
netstat -ano | findstr 27017
```

Expected:

```
LISTENING
```

Compass can now connect normally to:

```
mongodb://localhost:27017
```